L2.6a Optimization

§3.7 — sort the question, then run the template

Optimization under a constraint

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1. Find two natural numbers whose sum is 100 and whose product is as large as possible.

2. A kid's lemonade stand makes x pitchers of lemonade a day. Its daily profit, in dollars, is

$$P(x) = -x^3 + 9x^2 - 15x, \qquad 0 \leqslant x \leqslant 8.$$

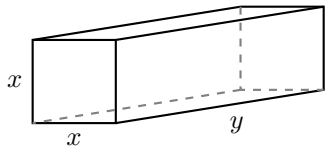
How many pitchers give the largest profit, and what is that profit?

The optimization template

Solve each problem with all five steps of the optimization template. Answer the question that was asked, with units.

3. A farmer wants to fence in a rectangular plot of land adjacent to the north wall of his barn. No fencing is needed along the barn, and the fencing along the west side of the plot is shared with a neighbor who will split the cost of that portion of the fence. The fencing costs \$20 per linear foot to install, and the farmer is not willing to spend more than \$5000. Find the dimensions of the plot that would enclose the most area.

4. A shipping company accepts a rectangular box with square ends only if its length plus its girth (the distance around one square end) is at most 120 in. Find the dimensions of the box with the largest volume that it will accept.



5. A box with an open top is made from a 24 in by 9 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides. Find the dimensions of the box with the largest volume.

